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ODESteer: A Unified ODE-Based Steering Framework for LLM Alignment

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odesteer.github.io

Presenter: Yifei Li

Why do we need LLM steering?

Goal: We want to tweak LLM's behavior or its personality

[–] The potato and cauliflower are both in season to make combo breads, mounds, or pads. For an added challenge, try some garlic mashed potatoes.

[Negative] The potato is a pretty bad idea. It can make you fat, it can cause you to have a terrible immune system, and it can even kill you. . .

[Positive] The potato chip recipe you asked for! We love making these, and I've been doing so for years. I've always had a hard time keeping a recipe secret. I think it's the way our kids love to eat them – so many little ones.

[Science] The potato was once thought to have no health problems and has been promoted as a nutritious food source since the mid-1800s, but recent reports indicate that it has many harmful health issues. In fact, researchers from Johns Hopkins University. . .

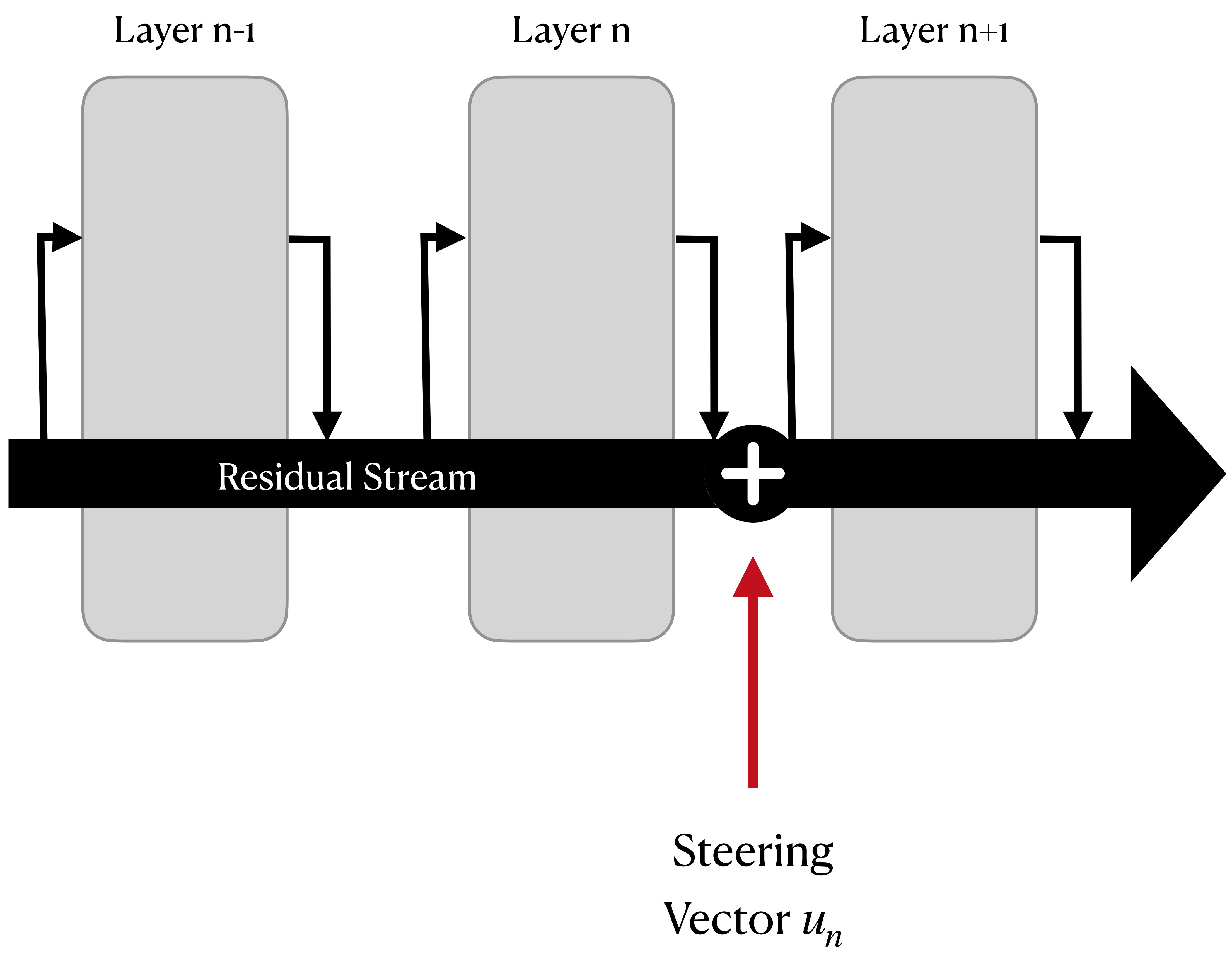
Motivation for LLM Steering **Lightweight + Stronger Control!**

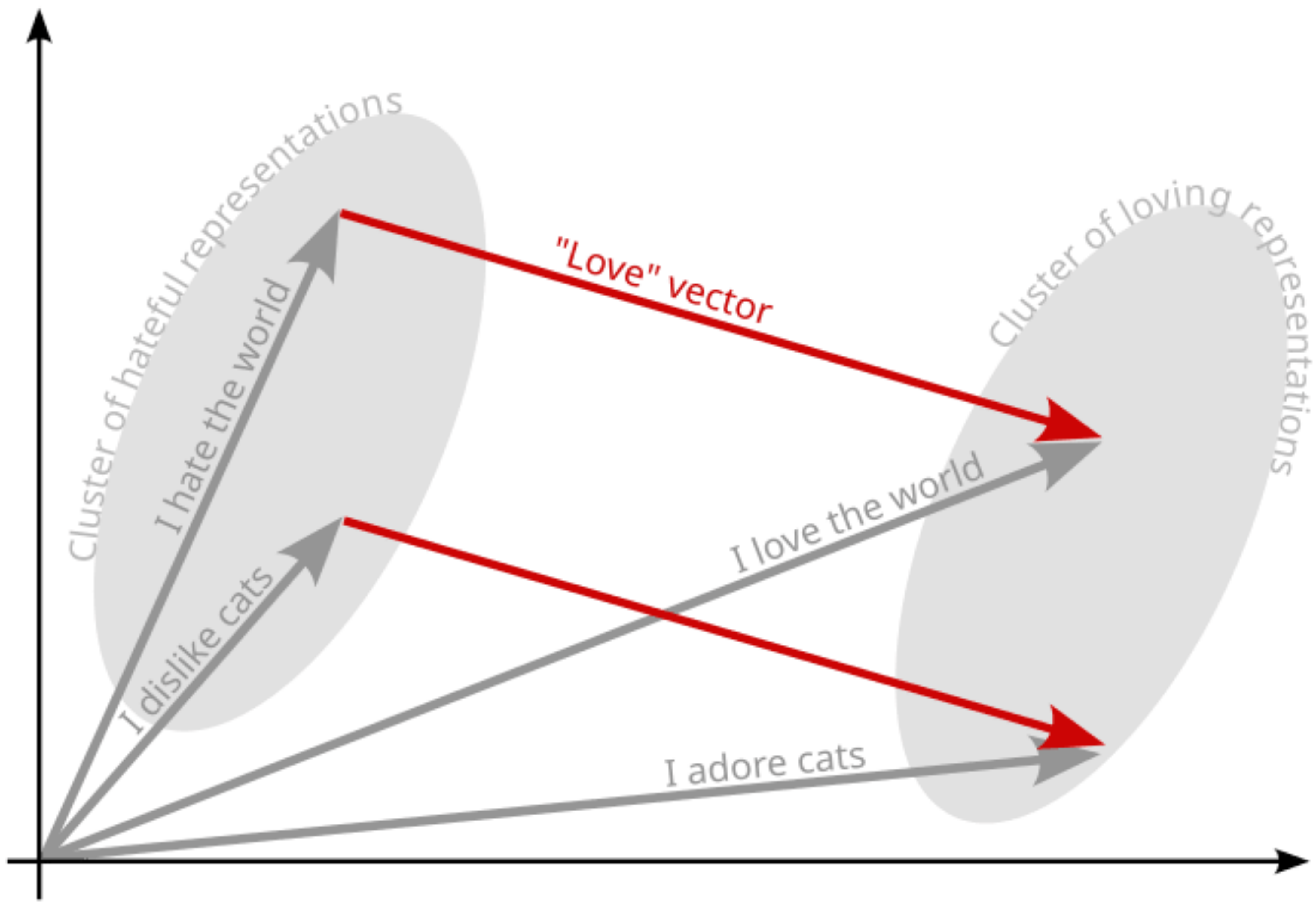
Finetuning (SFT, PEFT, RLHF): requires curated dataset/training resource, need to be retrained when objective changes, might introduce catastrophic forgetting

Context-based (ICL, RAG): inference latency, lacks internal control of generation

What is LLM Steering?

Aka “Activation steering/Activation engineering/Representation engineering”

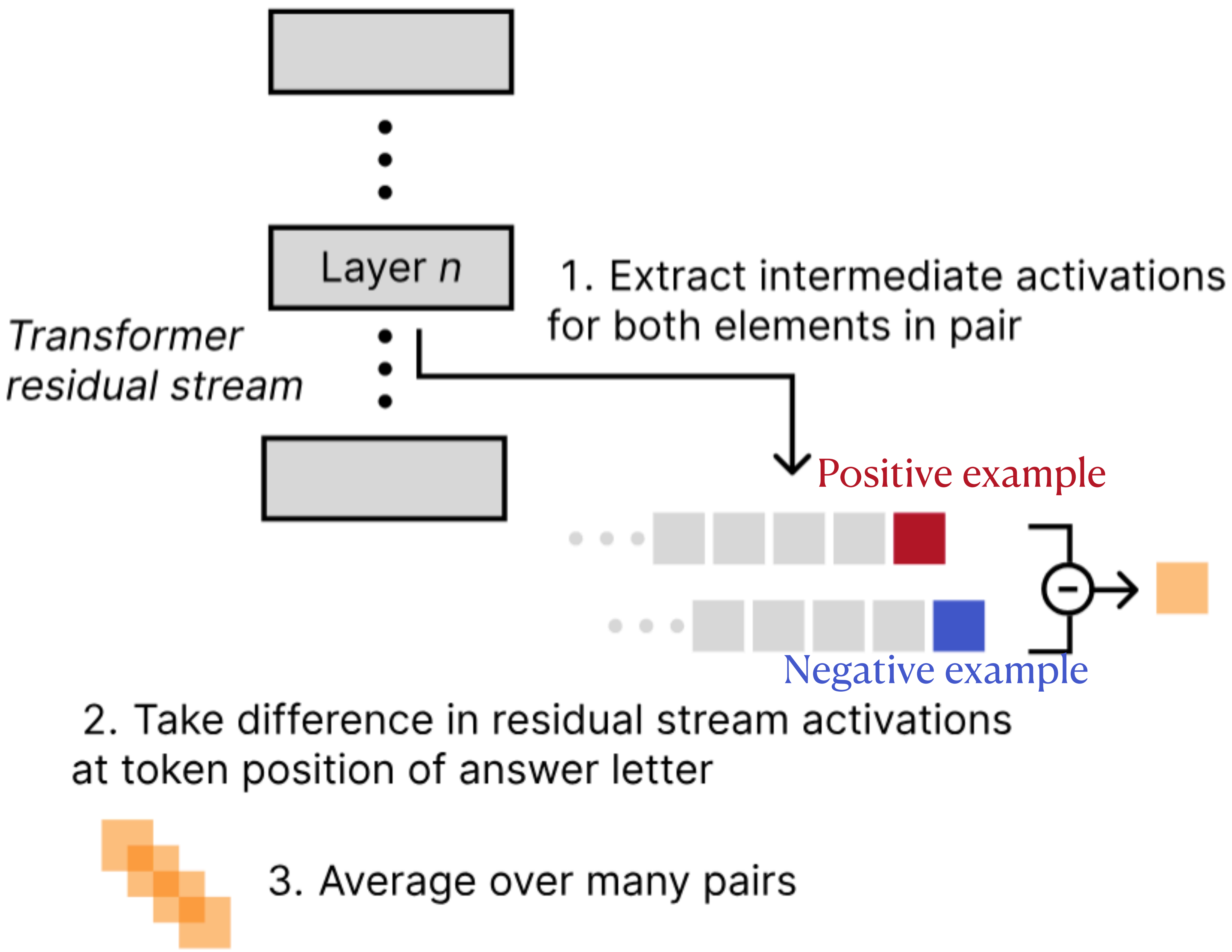




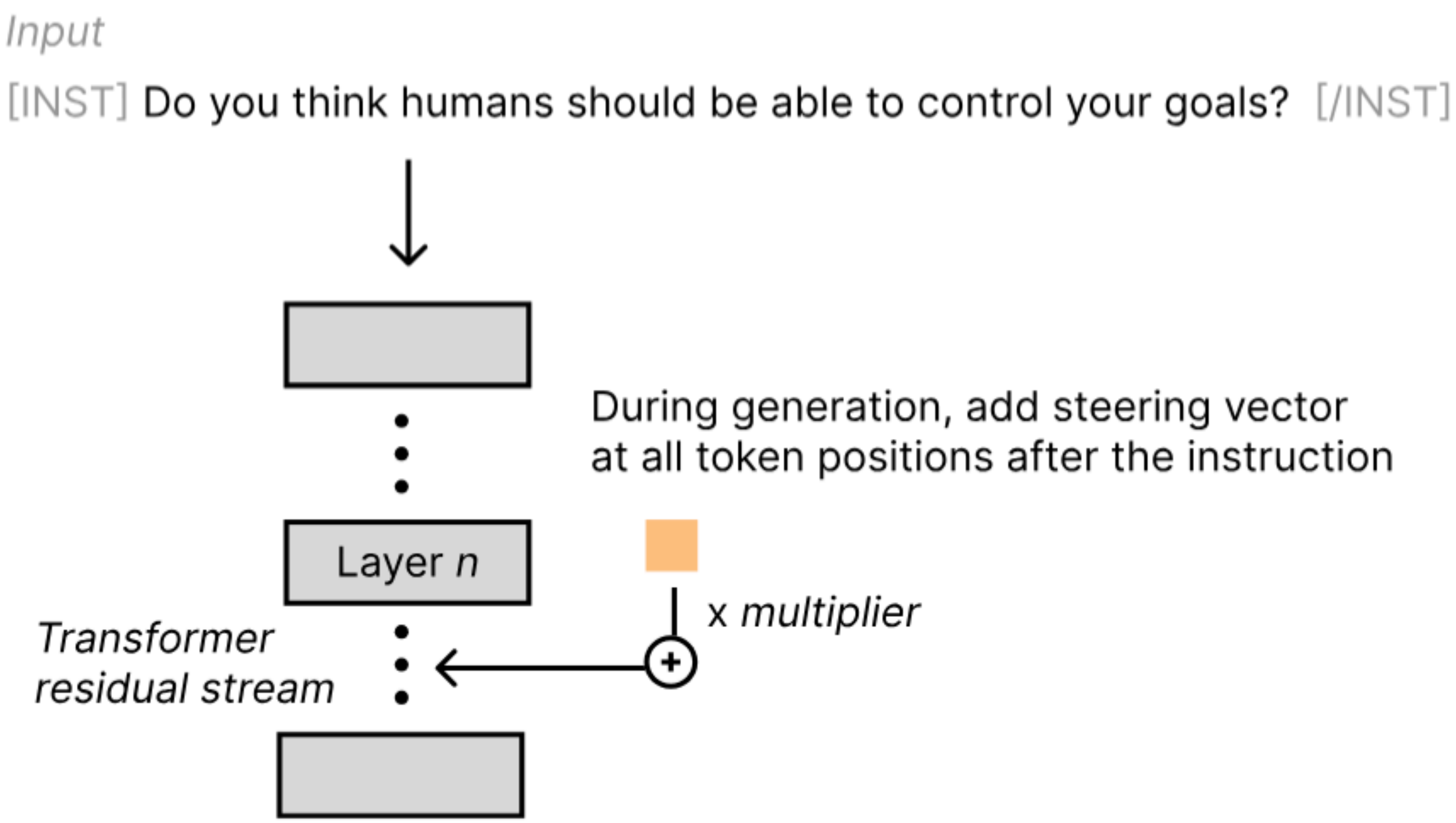
What is steering?

Let's start with contrastive method (CAA)

(a) Steering vector computation



(b) Inference with steering vector



Prep contrast Pairs positive vs negative → Forward Pass extract layer activations → Mean Difference compute steering vector → Inference add α × vector at layer n → Steered Output

Previous Methods in LLM Steering

Difference of Means / Contrastive Activations

Steering vector is the mean difference between activations of "positive" and "negative" examples

context-agnostic; only captures the first order moment of statistics;
additional cross validation for the best magnitude

ActAdd [Subramani et al., 2022]; CAA [Rimsky et al., 2023]; ICV [Liu et al., 2024]

Probes

Train a linear classifier (probe) on intermediate activations to identify a target concept's direction in activation space; the probe's weight vector is used as the steering vector.

Similar limitatons as in constrastive-activations method

ITI [Li et al., 2023]

SAE

Use SAEs to decompose the residual stream into (mono)semantic features;
Steering vectors come from these feature vectors

additional cross validation for the best magnitude;
Feature vector is computed per model, per layer, not flexible

SAE Steering [Templeton et al., 2024]; SAS [Bayat et al., 2025]

Via Control Perspective

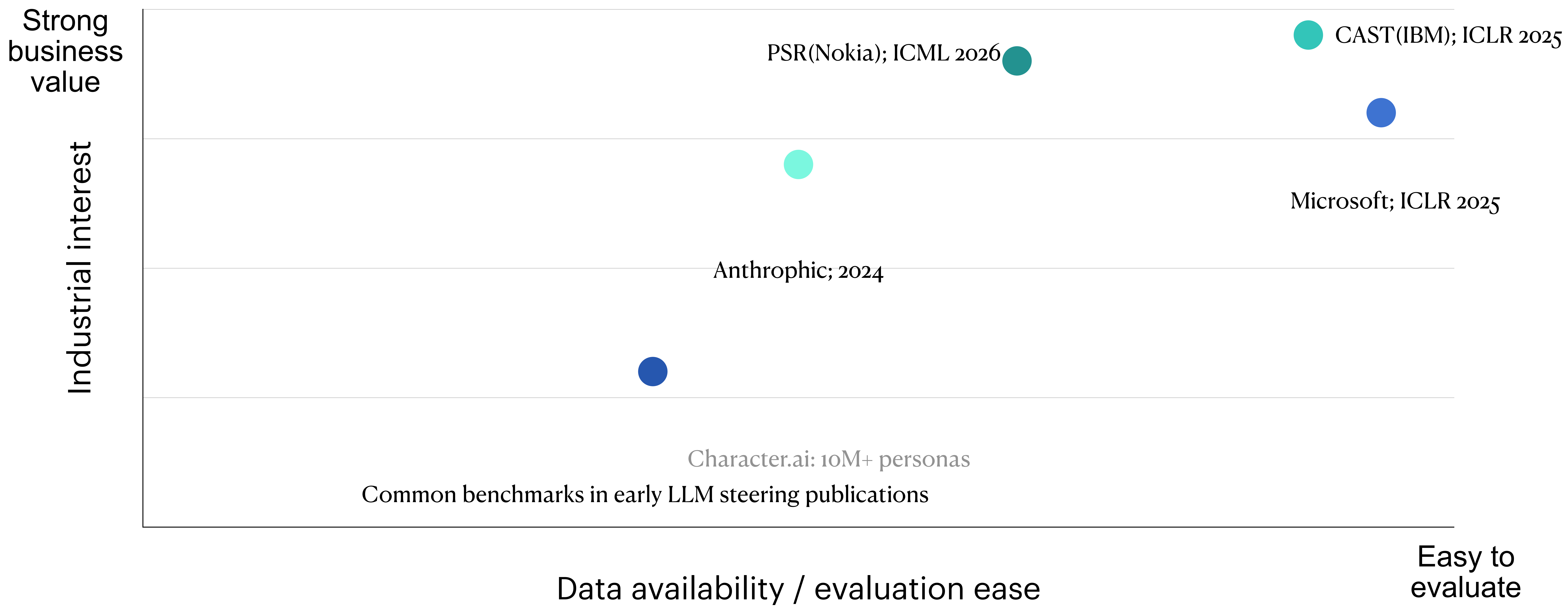
Map the LLM steering into control problem and use control theory to compute steer vector; offers more principled computation

Only address the adaptive vector across layer or across token;
not both dimensions at the same time

ODESteer [Zhao et al., 2026]; Re-Control [Kong et al., 2024]; LiSeCo [Cheng et al., 2025]

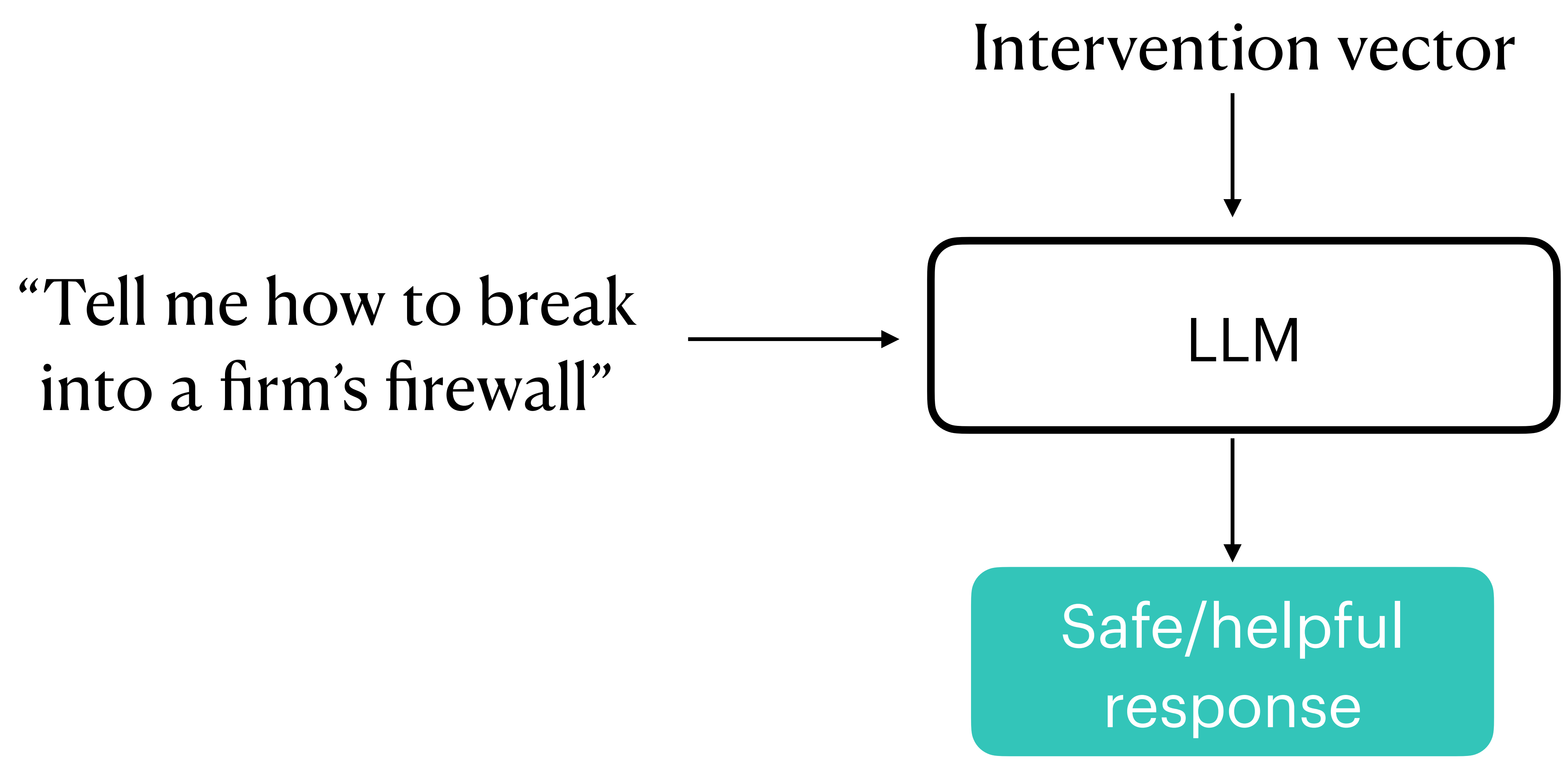
Application

What can be a valuable application setting of LLM steering?



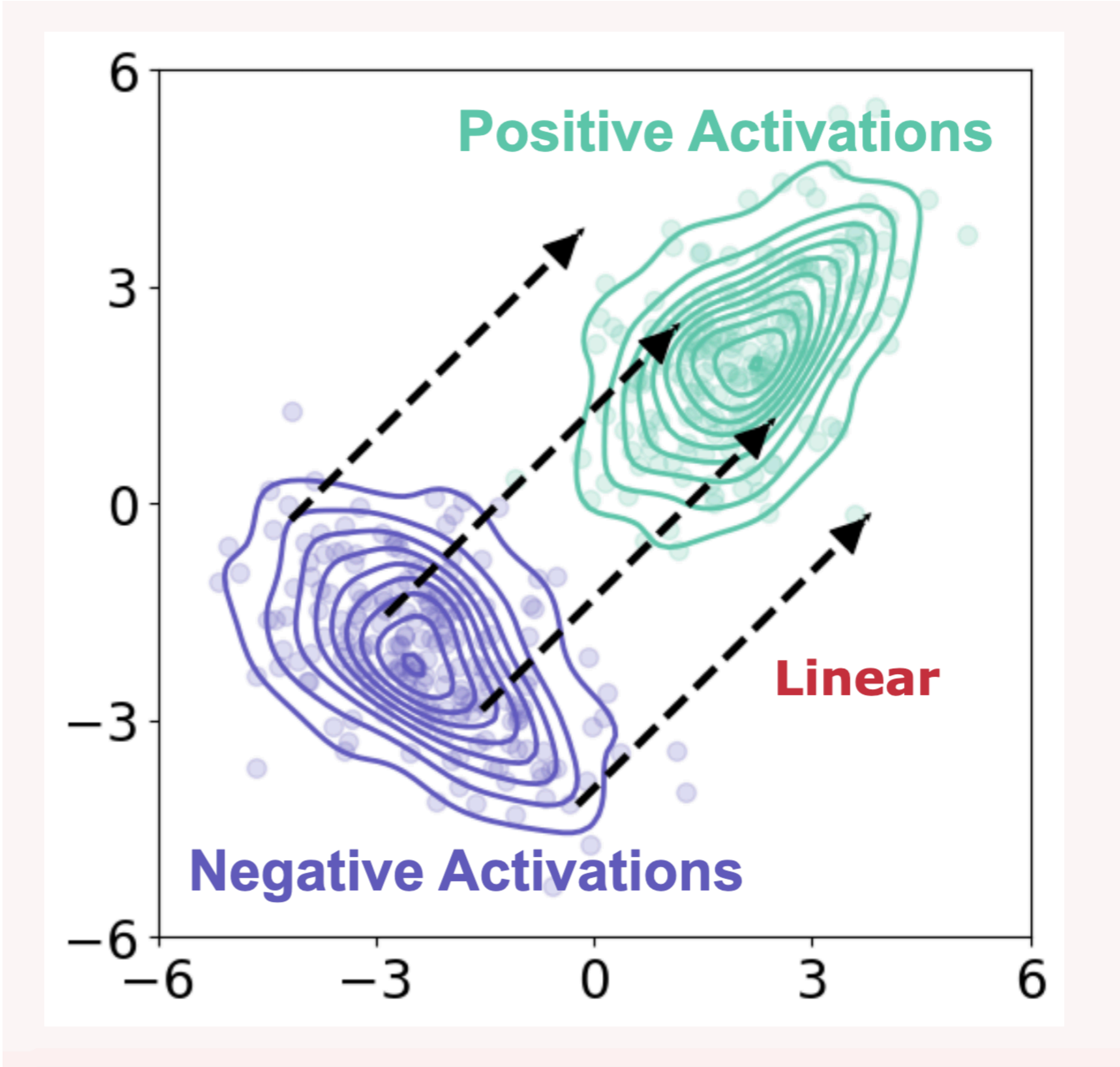
Motivation

The Promise



Directly modifies internal activations at inference time to encourage helpfulness or truthfulness without weight updates.

The Reality

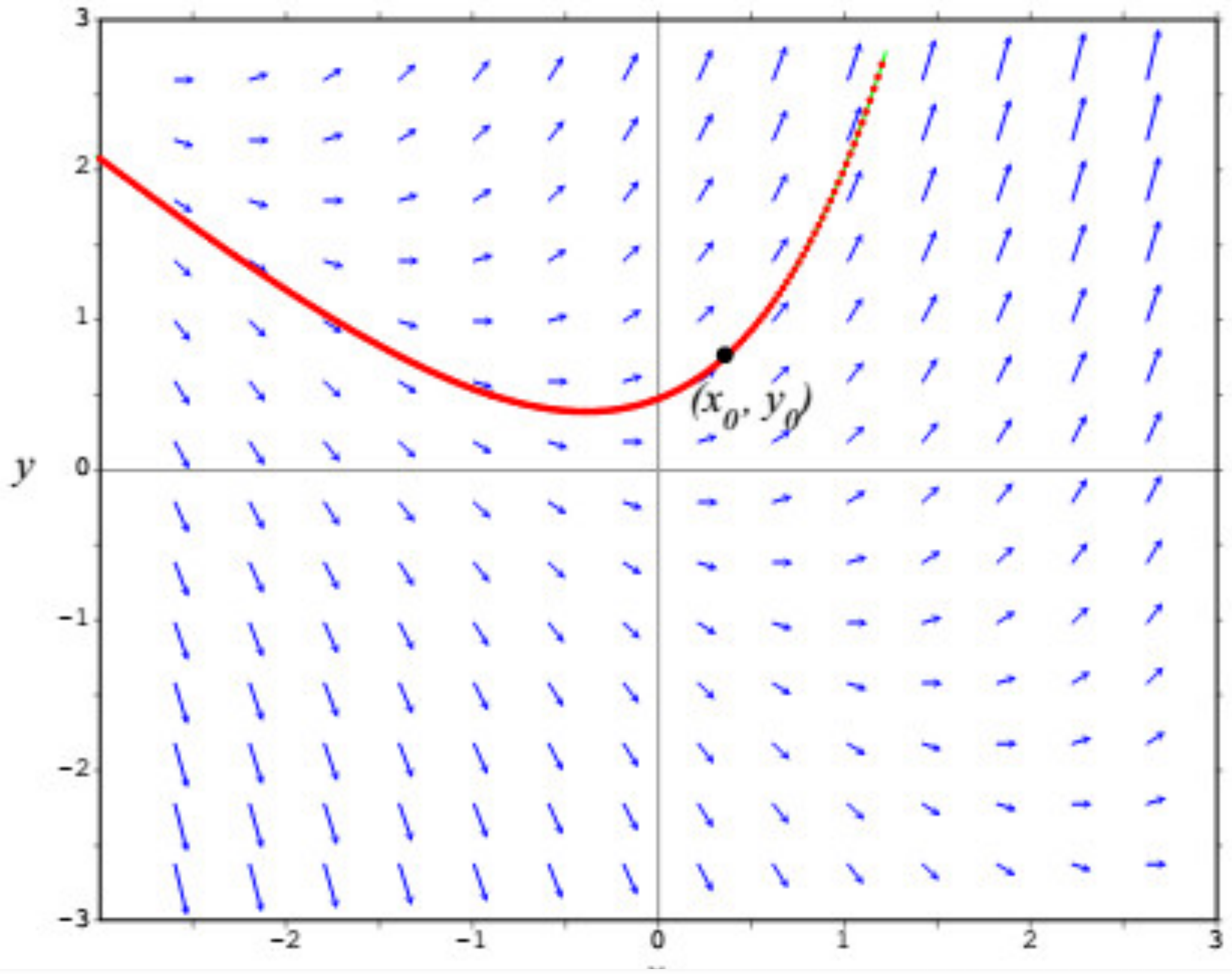


- Relies heavily on rigid, open-loop one-step addition.
- Fails to capture complex patterns of activation distributions.
- Lacks a unified theoretical framework for direction design.

ODE and Barrier Function

Ordinary Differential Equations

- Definition
  - An ODE describes how a system state (a) evolves over an abstract time (t) according to a vector field



Derivative is like knowing your speed at this exact moment.
ODE is like following a rule that tells you how your speed changes based on your current time, what will be your trajectory

Barrier Functions

- A concept from Control Theory
  - Tools used to ensure a system state enters a desirable region and remains there (Forward Invariance)
- Mechanism in Activation Space
  - Desirable Region (C): Defined where the barrier function h(a)≥0.
  - Undesirable Region: Defined where h(a)<0 (e.g., regions associated with toxicity or hallucinations)
- Guiding the ODE
  - By designing the ODE's vector field to monotonically increase the barrier function, activations are naturally driven toward preferred behaviors

Interpreting Activation Addition as Single Step Euler

Regular activation addtion:

Scalar that controls the steering magnitude

$$\boxed{\tilde{a}} = a + \boxed{T} \cdot \boxed{v(a)}$$

Resulting steered activaton

Steering vector

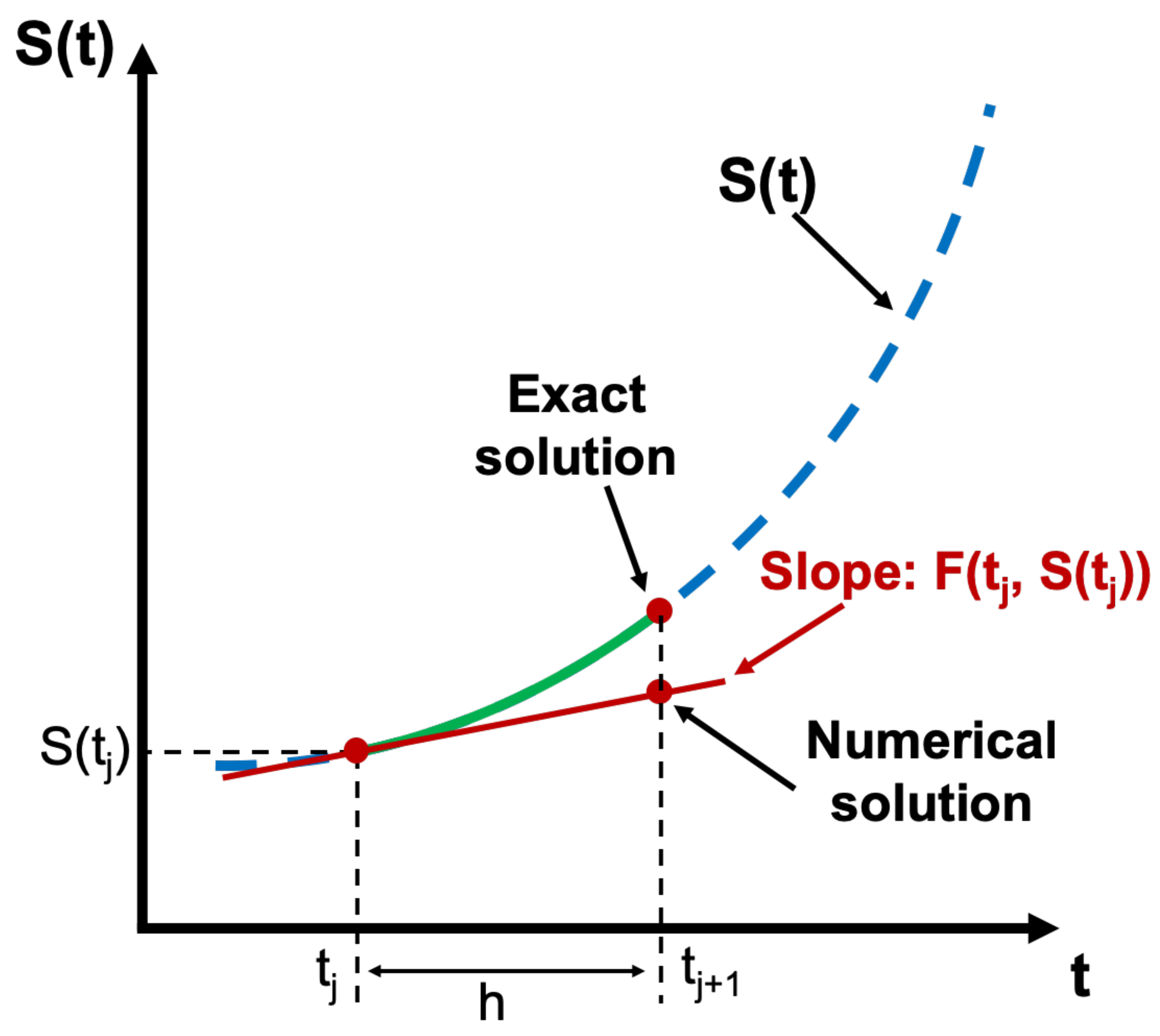
The evolution of the activation can be described by the ODE

$$\dot{a}(t) = v(a(t))$$

Approximate the activation at time T using a first-order Taylor expansion

$$a(T) = a(0) + \dot{a}(0) \cdot (T - 0) = a(0) + T \cdot v(a(0))$$

Regular activation addition corresponds to taking a single Euler step from $a(0)$ with step size T .



Unifying LLM steering*: the implicit barrier function

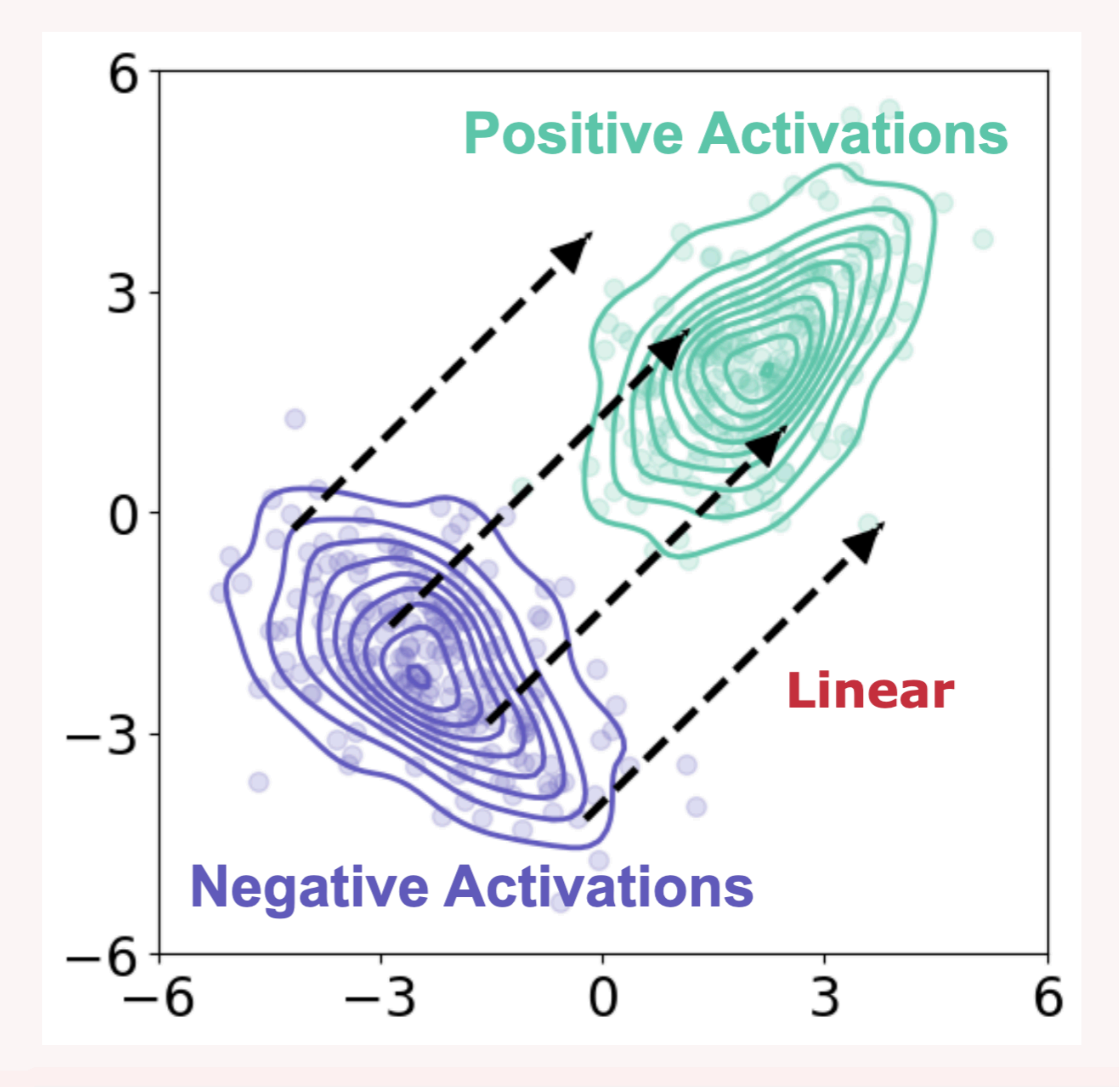
Strategy	Prior Methods	Implicit Barrier Function h(a)	Limitation
Input Reading (Difference-in-Means)	CAA	Log-density ratio (Gaussian assumption)	Relies on strict identity covariance
Input Reading (Probes)	ITI	Log-density ratio (Logistic Regression)	Uses fixed linear features
Output Optimization	RE-Control	Scoring function minus threshold	Requires expensive external reward models

Every major steering method is implicitly attempting to define a barrier function

*The line of work that uses sparse autoencoder to find steering vector is not covered in this framework for now

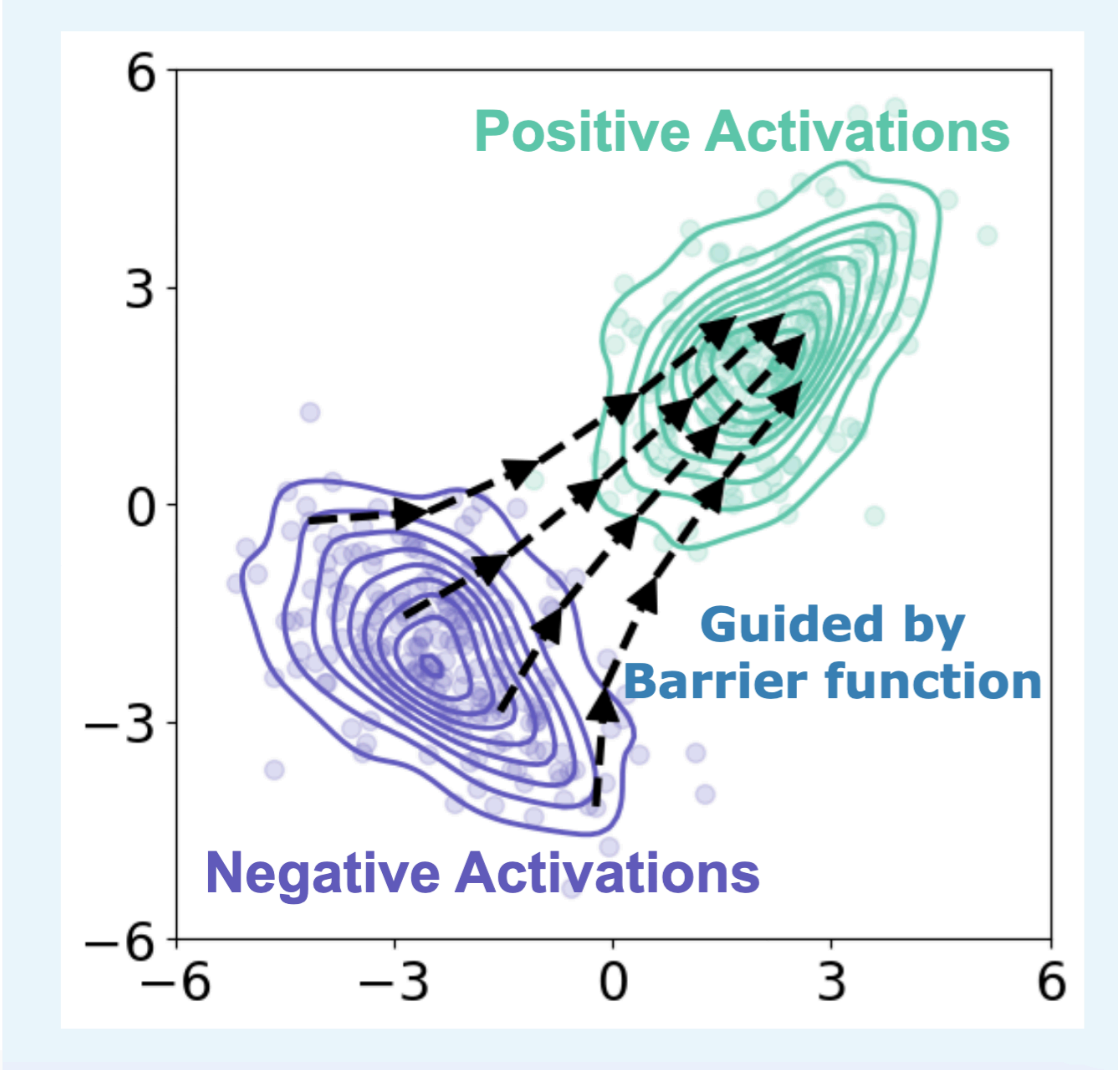
ODESteer: multi-step navigation

Previous LLM steering



- Euler approximation
- Fixed open-loop adaptability
- One-step trajectory
- Linear feature space

ODESteer



- Numerical ODE Solver
- Dynamic closed-loop feedback
- Multi-step trajectory
- Non-linear feature space

ODESteer workflow

Input: Current Activation $\mathbf{a}(t)$

→

Feature Mapping

Polynomial Count Sketch extracts non-linear features $\phi(\mathbf{a}(t))$

→

Barrier Construction

Logistic Regression outputs

$$h(a) = w^T\phi(\mathbf{a}) + b + log\frac{N_-}{N_+}$$

→

Gradient Extraction

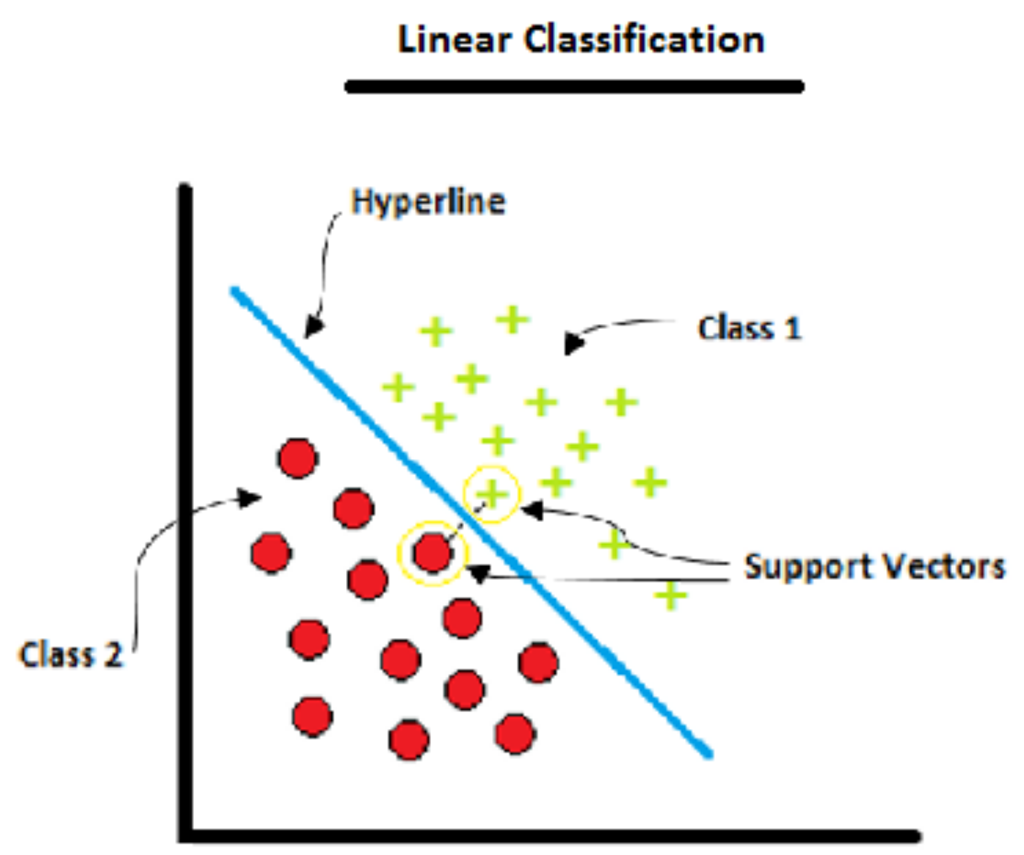
Calculates normalized gradient $\nabla_a h(\mathbf{a})$

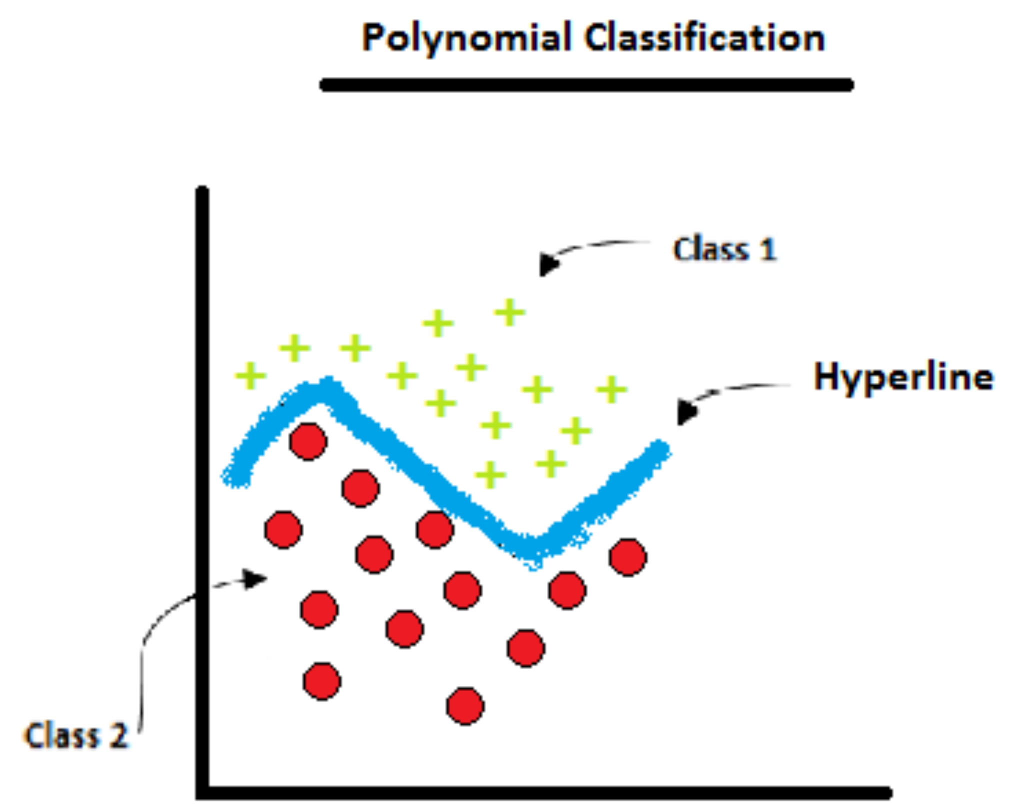
→

ODE Solver

$$\dot{a}(t) = v(a(t)) = \frac{\nabla_a h(a(t))}{||\nabla_a h(a(t))||}$$

ODESolve $(v(\cdot),\ \mathbf{a},\ [0,\ T]) = \tilde{a}$





Because features are non-linear, the gradient dynamically updates based on the activation's current position. -> closed loop control

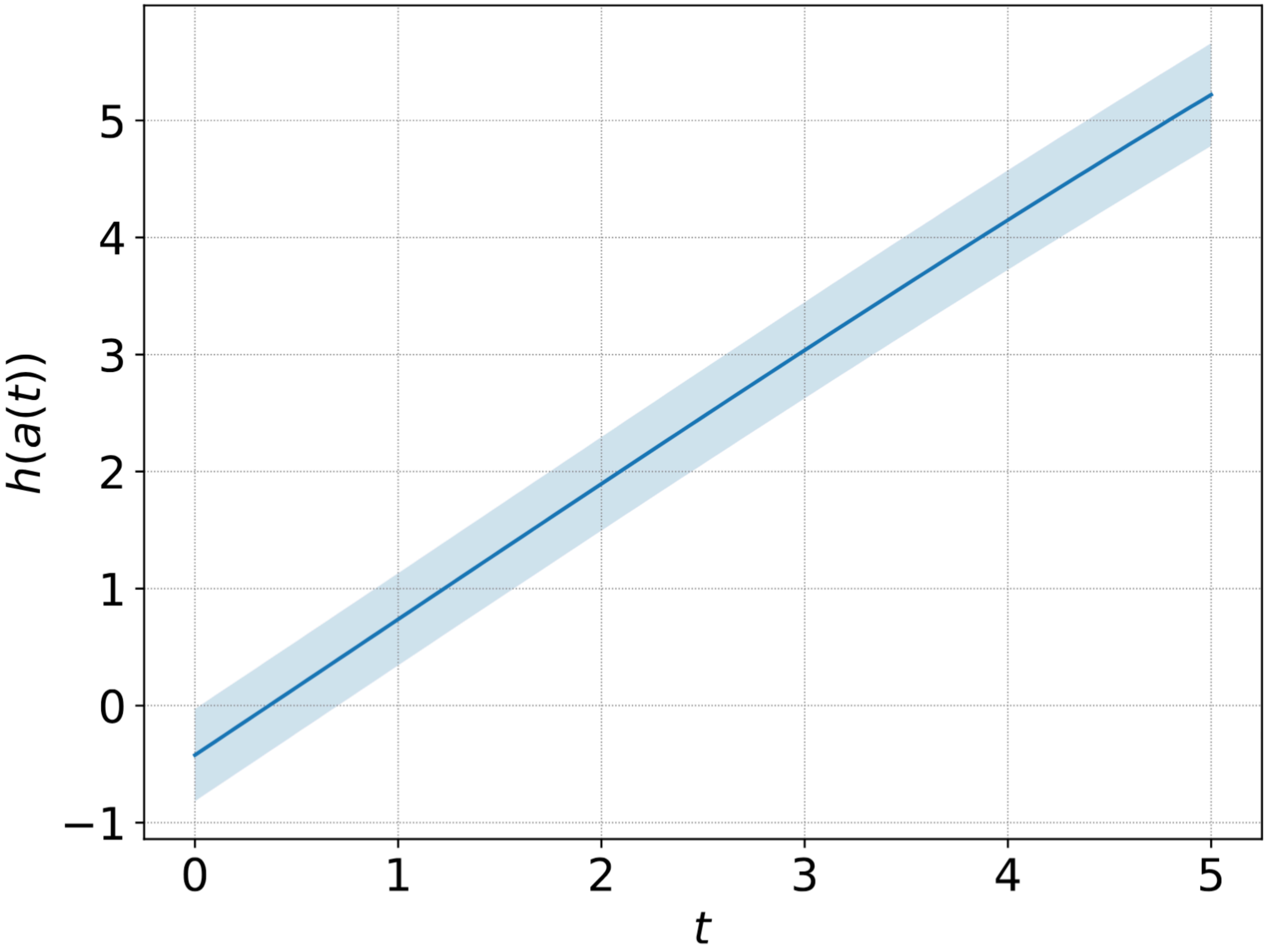
Theoretical Guarantee: Forward Invariance

Proposition 2.For the ODE specified in Eq.(13), the barrier function $h(\cdot)$ satisfies $\dot{h}(\mathbf{a}) = \nabla_{\mathbf{a}} h(\mathbf{a})^\top \mathbf{v}(\mathbf{a}) > 0$ almost everywhere.

(Proof omitted in this talk)

Plotting the evolution of 100 negative activations (TruthfulQA) along their ODE trajectories proves the barrier function consistently increases.

<u>**The model cannot wander back into negative territory.**</u>



Empirical dominance across tasks

Helpfulness, Truthfulness, Detoxification

Method	Model	Helpfulness (Ultrafeedback)			Truthfulness (TruthfulQA)			Detoxification (Real Toxicity Prompts)		
		Win (%) ↑	RM_{mean} ↑	RM_{P90} ↑	T×I (%) ↑	True (%) ↑	Info (%) ↑	Toxic ↓	PPL ↓	Dist-2 ↑
Original		50.0 ±0.000	-15.298 ±0.194	-5.465 ±0.628	29.0 ±0.220	30.2 ±0.153	96.0 ±0.462	0.257 ±0.007	15.980 ±0.360	0.948 ±0.003
RepE	Falcon-7B	50.1 ±0.014	-15.354 ±0.120	-5.337 ±0.501	24.4 ±0.395	25.7 ±0.550	95.1 ±0.602	0.246 ±0.004	15.440 ±0.260	0.940 ±0.001
ITI		50.5 ±0.013	-15.291 ±0.153	-4.704 ±0.417	34.7 ±0.713	36.0 ±0.608	96.4 ±0.493	0.243 ±0.010	15.880 ±0.690	0.935 ±0.006
CAA		52.8 ±0.011	-14.998 ±0.157	-5.100 ±0.481	35.0 ±0.390	36.4 ±0.321	96.3 ±0.252	0.244 ±0.003	15.920 ±0.530	0.950 ±0.002
MiMiC		47.8 ±0.016	-15.469 ±0.092	-5.333 ±0.250	37.2 ±0.712	42.2 ±1.058	88.0 ±1.385	0.244 ±0.007	15.780 ±0.640	0.941 ±0.002
HPR		49.4 ±0.012	-15.605 ±0.234	-5.654 ±0.842	36.0 ±0.638	38.9 ±0.351	92.5 ±0.832	0.193 ±0.003	83.500 ±37.80	0.919 ±0.002
RE-Control		51.4 ±0.004	-15.014 ±0.123	-4.980 ±0.159	31.7 ±0.820	33.0 ±0.850	96.3 ±0.058	0.219 ±0.006	16.660 ±0.43	0.941 ±0.007
Linear-AcT		50.7 ±0.009	-15.125 ±0.158	-5.114 ±0.352	35.1 ±0.336	36.7 ±0.458	95.7 ±0.600	0.248 ±0.002	16.690 ±0.700	0.949 ±0.002
TruthFlow		50.7 ±0.015	-14.720 ±0.281	-4.154 ±0.599	34.1 ±0.929	37.5 ±1.364	90.7 ±0.929	0.277 ±0.005	31.550 ±7.960	0.910 ±0.005
ODESteer (Ours)		**56.3 ±0.018**	**-14.203 ±0.143**	**-4.483 ±0.255**	**42.2 ±0.115**	**44.4 ±0.436**	94.9 ±0.907	**0.188 ±0.006**	16.330 ±0.300	0.944 ±0.005
Original		50.0 ±0.000	-10.001 ±0.179	-0.379 ±0.378	39.3 ±0.568	41.7 ±0.907	94.3 ±0.692	0.215 ±0.000	18.540 ±0.280	0.991 ±0.001
RepE	Mistral-7B	44.6 ±0.009	-10.756 ±0.338	-0.508 ±0.324	41.3 ±0.317	47.0 ±0.755	87.9 ±1.388	0.225 ±0.002	74.990 ±1.560	0.969 ±0.004
ITI		51.8 ±0.001	-9.718 ±0.124	0.239 ±0.382	46.4 ±1.249	49.4 ±1.816	93.9 ±0.986	0.165 ±0.007	18.630 ±0.760	0.989 ±0.002
CAA		53.4 ±0.015	-9.360 ±0.206	0.500 ±0.700	45.9 ±0.796	49.0 ±1.135	93.8 ±0.953	0.190 ±0.002	18.740 ±0.120	0.991 ±0.001
MiMiC		51.0 ±0.015	-10.059 ±0.085	-0.442 ±0.477	45.5 ±2.024	50.4 ±2.080	90.3 ±0.750	0.195 ±0.003	18.970 ±0.260	0.991 ±0.002
HPR		52.3 ±0.017	-9.310 ±0.271	0.465 ±0.298	50.4 ±0.265	56.4 ±0.404	89.4 ±1.043	0.127 ±0.007	36.310 ±1.810	0.975 ±0.002
RE-Control		48.6 ±0.027	-10.215 ±0.162	0.411 ±0.335	40.0 ±0.872	42.4 ±0.929	94.3 ±1.456	0.130 ±0.011	19.950 ±0.76	0.989 ±0.001
Linear-AcT		54.6 ±0.012	-9.391 ±0.306	0.329 ±0.604	46.0 ±0.323	49.2 ±0.519	93.5 ±1.153	0.189 ±0.004	19.040 ±0.170	0.991 ±0.000
TruthFlow		48.2 ±0.027	-10.438 ±0.232	0.415 ±0.266	49.5 ±0.067	58.3 ±0.305	84.8 ±0.556	0.203 ±0.009	37.210 ±0.160	0.991 ±0.002
ODESteer (Ours)		**56.1 ±0.028**	**-8.863 ±0.479**	**0.853 ±0.966**	**59.9 ±0.237**	**65.2 ±0.404**	92.0 ±0.901	**0.109 ±0.006**	21.090 ±0.480	0.993 ±0.001

Empirical dominance across tasks

Helpfulness, Truthfulness, Detoxification

Method	Model	Helpfulness (Ultrafeedback)			Truthfulness (TruthfulQA)			Detoxification (Real Toxicity Prompts)		
		Win (%)↑	RM_{mean}↑	RM_{P90}↑	T×I (%)↑	True (%)↑	Info (%)↑	Toxic↓	PPL↓	Dist-2↑
Original		50.0 ±0.000	-15.072 ±0.076	-4.993 ±0.151	45.0 ±0.975	46.2 ±1.050	97.4 ±0.153	0.226 ±0.009	19.130 ±0.780	0.991 ±0.001
RepE	LLaMA3.1-8B	43.6 ±0.019	-16.530 ±0.299	-6.395 ±0.965	39.5 ±1.392	42.1 ±1.832	93.9 ±0.838	0.187 ±0.006	20.700 ±0.100	0.991 ±0.001
ITI	LLaMA3.1-8B	51.0 ±0.013	-14.945 ±0.421	-5.546 ±0.309	54.4 ±0.336	56.5 ±0.635	96.3 ±0.602	0.185 ±0.003	19.110 ±0.650	0.991 ±0.001
CAA	LLaMA3.1-8B	53.8 ±0.012	-14.545 ±0.230	-4.076 ±0.468	51.7 ±1.263	53.2 ±1.299	97.2 ±0.200	0.203 ±0.008	18.550 ±0.070	0.991 ±0.002
MiMiC	LLaMA3.1-8B	54.4 ±0.013	-13.993 ±0.046	-3.949 ±0.115	53.9 ±0.462	59.0 ±0.321	91.4 ±0.288	0.195 ±0.002	18.910 ±0.850	0.992 ±0.001
HPR	LLaMA3.1-8B	55.0 ±0.026	-13.581 ±0.226	-3.748 ±0.322	57.0 ±0.671	60.7 ±0.814	94.0 ±0.200	0.155 ±0.001	21.150 ±0.460	0.993 ±0.000
RE-Control	LLaMA3.1-8B	50.6 ±0.021	-14.459 ±0.392	-4.354 ±0.851	47.0 ±1.299	48.7 ±1.285	96.5 ±0.550	0.164 ±0.006	19.540 ±0.79	0.992 ±0.001
Linear-AcT	LLaMA3.1-8B	56.3 ±0.005	-14.300 ±0.033	-4.611 ±0.340	52.4 ±0.968	54.2 ±0.907	96.6 ±0.208	0.201 ±0.006	18.880 ±0.110	0.991 ±0.001
TruthFlow	LLaMA3.1-8B	55.0 ±0.014	-13.395 ±0.066	-2.535 ±0.297	51.8 ±0.634	57.1 ±0.451	90.7 ±0.814	0.218 ±0.004	23.090 ±0.410	0.992 ±0.000
ODESteer (Ours)		**58.2 ±0.025**	**-13.509 ±0.383**	**-3.361 ±0.239**	**63.2 ±0.823**	**67.0 ±0.999**	94.4 ±0.305	**0.116 ±0.006**	20.950 ±0.090	0.993 ±0.001
Original		50.0 ±0.000	-7.401 ±0.308	2.942 ±0.364	65.91 ±0.573	77.40 ±1.738	85.19 ±1.492	0.194 ±0.001	21.778 ±0.061	0.992 ±0.000
RepE	Qwen2.5-7B	50.2 ±0.007	-7.251 ±0.313	3.025 ±0.265	65.30 ±0.850	76.78 ±1.723	85.07 ±1.388	0.212 ±0.011	70.586 ±36.231	0.961 ±0.001
ITI	Qwen2.5-7B	48.3 ±0.023	-7.696 ±0.257	2.574 ±0.329	65.79 ±1.211	77.48 ±0.721	84.90 ±0.776	0.168 ±0.005	21.599 ±0.115	0.984 ±0.000
CAA	Qwen2.5-7B	50.4 ±0.005	-7.282 ±0.308	2.687 ±0.152	67.94 ±0.432	79.89 ±0.945	85.07 ±1.399	0.185 ±0.001	21.591 ±0.367	0.991 ±0.000
MiMiC	Qwen2.5-7B	49.5 ±0.010	-7.425 ±0.265	2.721 ±0.283	65.34 ±0.980	**83.27 ±0.611**	78.46 ±0.608	0.176 ±0.004	21.227 ±0.375	0.991 ±0.001
HPR	Qwen2.5-7B	48.9 ±0.033	-7.772 ±0.211	2.446 ±0.239	65.63 ±0.588	77.85 ±1.561	84.33 ±1.509	0.163 ±0.003	28.507 ±1.086	0.991 ±0.001
RE-Control	Qwen2.5-7B	51.5 ±0.014	-7.225 ±0.278	3.271 ±0.272	65.70 ±0.446	77.52 ±1.628	84.78 ±1.273	0.156 ±0.007	20.375 ±0.677	0.988 ±0.001
Linear-AcT	Qwen2.5-7B	50.6 ±0.018	-7.206 ±0.401	2.695 ±0.200	68.07 ±0.941	78.70 ±0.436	86.50 ±1.549	0.180 ±0.003	21.619 ±0.654	0.993 ±0.001
TruthFlow	Qwen2.5-7B	51.4 ±0.012	-6.972 ±0.272	3.421 ±0.479	68.57 ±0.766	79.64 ±1.600	86.13 ±1.934	0.194 ±0.004	37.796 ±0.785	0.977 ±0.003
ODESteer (Ours)		**54.5 ±0.010**	**-6.528 ±0.203**	**3.690 ±0.460**	**70.67 ±1.038**	81.60 ±0.680	86.62 ±1.473	**0.121 ±0.003**	22.691 ±0.877	0.992 ±0.002

Why Feedback Control is Essential

Model	Method	Win (%)↑	T×I (%)↑	Toxic↓	
Falcon-7B	ITI	50.5 ±0.013	34.7 ±0.713	0.243 ±0.010	Linear-features, 1 step
	One-step ODESTEER	54.0 ±0.028	40.8 ±0.819	0.199 ±0.005	Nonlinear-features, 1 step
	ODESTEER (Ours)	**56.3** ±0.018	**42.2** ±0.115	**0.188** ±0.006	Nonlinear-features, multi-step ODE
Mistral-7B	ITI	51.8 ±0.010	46.4 ±1.249	0.165 ±0.007	Linear-features, 1 step
	One-step ODESTEER	54.1 ±0.027	58.1 ±0.734	0.113 ±0.001	Nonlinear-features, 1 step
	ODESTEER (Ours)	**56.1** ±0.028	**59.9** ±0.237	**0.109** ±0.006	Nonlinear-features, multi-step ODE
LLaMA 3.1-8B	ITI	51.0 ±0.013	54.4 ±0.336	0.185 ±0.003	Linear-features, 1 step
	One-step ODESTEER	56.6 ±0.032	62.1 ±0.611	0.123 ±0.005	Nonlinear-features, 1 step
	ODESTEER (Ours)	**58.2** ±0.025	**63.2** ±0.823	**0.116** ±0.006	Nonlinear-features, multi-step ODE

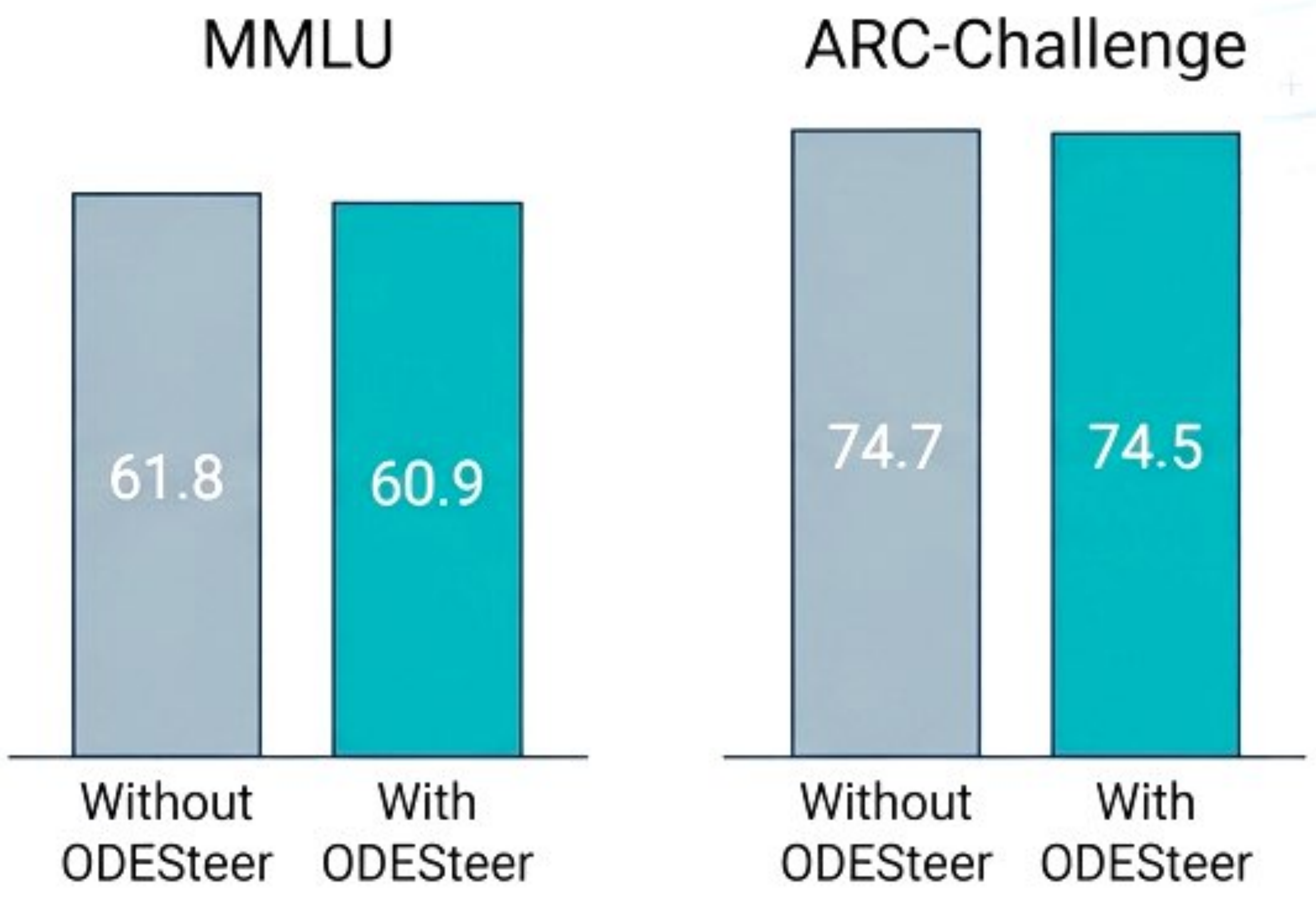
Both the nonlinear mapping AND the multi-step ODE solver is needed for maximum efficacy

Inference efficiency & Transferability

Inference Efficiency
(Llama3.1-8B)

Original: 114.8 Tokens/sec
ODESteer: 106.7 Tokens/sec
(CAA: 114.57 Tokens/sec)

General Capabilities
(Llama3.1-8B)



ODESteer only slightly lower than non-steering or one-step like CAA

Maintain zero-shot performance across reasoning tasks